

Who Works for a House?

Service employment across the residential spectrum — from a local worker's studio to a 15,000-square-foot second home used one month a year. A statistical companion to Housing in the Upper San Miguel.

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1. What this report answers

Every home in the Region creates paid, year-round work: someone plows the driveway, checks the boiler, cleans, landscapes, and manages the property. Impact-fee studies have tried to measure this for twenty-five years, and their central claim is right even where their arithmetic is not. This report rebuilds the number from the two independent surveys that actually measured it, states every assumption, and answers a specific question: **how much full-time-equivalent (FTE) employment does each kind of home generate — from a 500-square-foot studio occupied by a local worker to a 15,000-square-foot second home occupied about one month a year?**

The headline findings:

- **Home size drives service employment on a measured curve.** Employment scales as roughly size to the 1.6 power — our independent refit of the 2000 four-county survey gives an exponent of 1.57; the 2013 Teton County regression, on fresh data, found 1.59.¹² Two surveys, twelve years apart, two states, one curve.
- **A local worker's studio generates about 0.01 FTE of paid property service. A 15,000-square-foot second home used one month a year generates roughly 0.5–2.0 FTE — centrally, about one full-time worker, every year, indefinitely.**³
- **The labor balance is the useful statistic.** The studio houses a worker who fills about 1.2 local jobs; the big house houses no one and consumes about one worker. **One month-a-year mansion cancels the workforce contribution of one occupied studio.**
- **Construction employment is real but it is not Telluride housing demand.** The construction workforce lives overwhelmingly down-valley and in Montrose and commutes — often in van pools — with no intention of relocating; that was already true in the 2000 survey data and remains true by observation today.⁴ It is a regional labor flow that loads Highway 145 and Society Turn, not the local housing queue.

2. The statistical rules this report follows

Five practices separate a defensible estimate from an advocacy number, and each one corrects a specific failure in the fee studies this report supersedes:

- **Measure the curve; never assume its shape.** The 2005 fee study fitted an exponential to six data points and extrapolated it to 14,000 sq ft — predicting 6.4 FTE where the largest measurement ever taken was 1.10, and 8.8 FTE at 15,000 sq ft.⁵ We fit the exponent from the data (a power law in logs), and both datasets independently agree on it.
- **Never extrapolate past the data without saying so.** Both source surveys top out near 8,000–9,000 sq ft; the 2013 authors refused to predict beyond 7,000 for exactly this reason.² Our 15,000 sq ft figures are flagged extrapolations of a measured curve — shown as a range, not a point.
- **Use ranges spanning the independent estimates.** The two surveys differ in level (the 2000 survey runs higher; its sample deliberately excluded local homes under 3,000 sq ft and over-represented the second-home market⁶). Rather than pick a winner, the low bound is the 2013 Teton calibration and the high bound is the 2000 survey at San Miguel County’s measured level.⁷
- **Prefer administrative data over surveys where it exists.** For construction, payroll records (ES-202) divided by permitted square footage give 1.23 FTE-years per 1,000 sq ft; contractor-survey methods gave 2.3–4.4. We carry the range and note that the administrative number is the anchor.⁸
- **Separate the channels and close the loop.** Property upkeep, resident spending, and construction are different economic objects with different housing implications; blending them is how the old studies both overcharged studios and undercharged estates. And every housed worker generates a small, convergent second round of demand — the loop multiplies need by about 1.10–1.35 and then stops.⁹

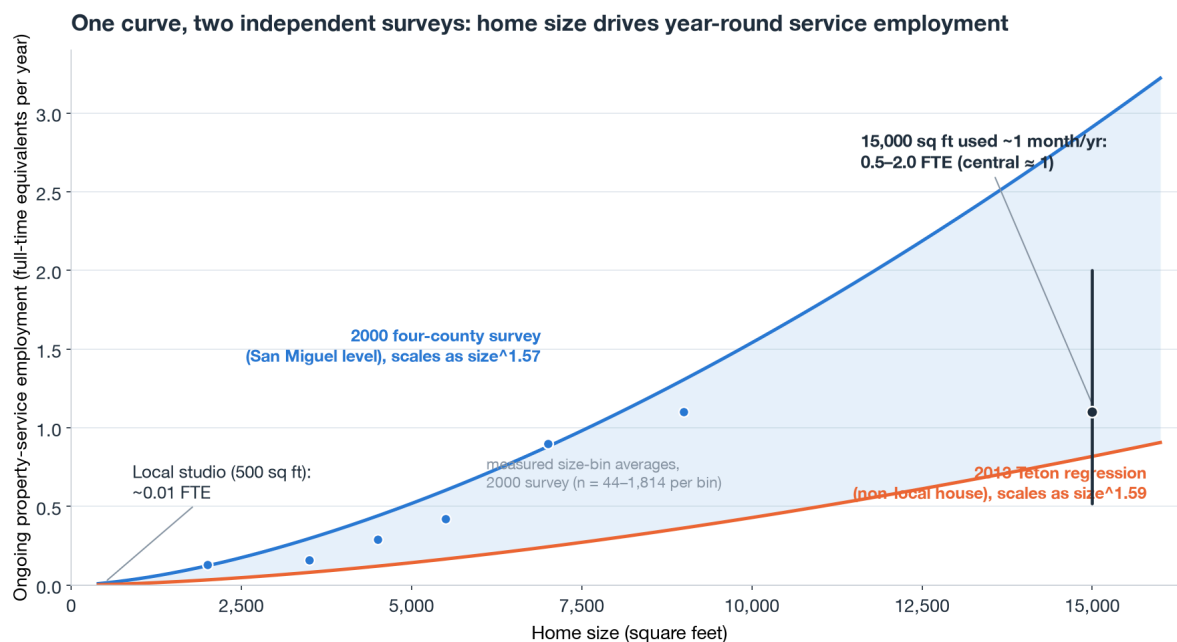
3. The evidence base

The 2000 Residential Job Generation Study (RRC Associates / The Housing Collaborative) surveyed 2,792 homeowners and 17 property-management firms across Teton County WY, Gunnison County, Breckenridge/Upper Blue, and San Miguel County — a 35% response rate, margin of error ± 1.85 points. It measured total ongoing employment of **0.23 FTE per unit overall and 0.29 in San Miguel County**, rising from 0.13 below 3,000 sq ft to 1.10 above 8,000.¹ Its known biases run upward: local primary homes under 3,000 sq ft were excluded by design, and its San Miguel construction sample was twelve custom homes averaging 7,784 sq ft.⁶

The 2013 Employee Generation by Land Use Study (Clarion Associates / Dr. James C. Nicholas / RRC, for Teton County) fielded a fresh 978-response homeowner survey and

estimated a log-linear regression of employment on size, tenancy, and unit type. All variables are strongly significant; size is by far the dominant one; local ownership cuts employment by about half and single-family form by more than half relative to condominiums at the same size. The model explains a third of household-level variance ($R^2 = 0.32$) — appropriate for setting averages, not individual predictions — and its authors capped its use at 7,000 sq ft.²

The two surveys disagree on level (0.23 vs 0.134 FTE per average unit) for knowable reasons — sample design, twelve years of professionalization of property management, different local economies — and agree almost exactly on shape. That is precisely the pattern one expects when the underlying relationship is real.⁷



Sources: Residential Job Generation Study (RRC/Housing Collaborative, 2000), Table II-13; Employee Generation by Land Use Study (Clarion/Nicholas/RRC, Teton County, 2013), Table 13. Band = the two calibrations; the month-a-year point applies the vacancy-intensity adjustment (endnote 8).

Exhibit A. The size–employment curve. Points are the 2000 survey’s measured size-bin averages; the blue curve is that survey refit as a power law at San Miguel County’s level; the orange curve is the 2013 Teton regression for a non-locally-owned house. The black range applies the month-a-year occupancy adjustment at 15,000 sq ft.

4. The ladder: FTE by product, studio to estate

The table reads in two channels. **Property service** is paid upkeep employment from the fitted curves (low = Teton 2013 calibration; high = 2000 survey at the San Miguel level). **Local spending** is the demand-side employment supported by the occupants’ purchases — near the national 0.30 jobs per occupied unit for year-round households, discounted for down-valley and online leakage; a month-a-year home contributes only an episodic sliver.¹⁰

Home	Size (sq ft)	Property service (FTE/yr)	Local spending (jobs/yr)	Total local jobs (FTE/yr)
Local worker studio (rental)	500	0.007–0.014	0.15–0.30	0.16–0.31
Local 1-bedroom condo	1,000	0.02–0.04	0.15–0.30	0.17–0.34
Local family home	2,000	0.02–0.12	0.20–0.30	0.22–0.42
Local larger home	3,500	0.06–0.30	0.20–0.30	0.26–0.60
Non-local home, typical use (~25 wks)	3,000	0.06–0.23	0.03–0.08	0.09–0.31
Second home, typical use (~25 wks)	6,000	0.19–0.68	0.04–0.10	0.23–0.78
Second home, typical use (~25 wks)	10,000	0.43–1.51	0.05–0.12	0.48–1.63
Second home used ~1 month/yr	15,000	0.5–2.0	0.01–0.05	0.5–2.0

Three features of the ladder deserve emphasis. First, **the paid-service channel spans two full orders of magnitude** — 0.01 FTE to roughly 1–2 FTE — across the residential spectrum; no flat per-unit rule can describe it. Second, for modest occupied housing the spending channel dominates the upkeep channel: a studio’s occupant supports far more local employment through groceries and services than through property maintenance — and the old nexus studies counted only the smaller channel. Third, deed-restricted local units sit at the very bottom of the generation curve, because local residents do their own upkeep — the 2013 regression measures local ownership at roughly half the employment of non-local at identical size.²

5. The month-a-year adjustment

Intuition says a home used four weeks a year should generate a twelfth of the employment of a year-round one. The data say otherwise. The 2000 survey measured employee hours both when owners are present and when they are absent: housekeepers work 34% as many hours in the owner’s absence, gardeners 35%, caretakers 62%.¹¹ An unoccupied mountain home is not dormant — it is heated, plowed, monitored, landscaped, repaired, and readied, year-round, at roughly **40% of occupied-week intensity**.

Scaling from the average second home (used 25.5 weeks, vacant 26.5¹²) to a four-week home therefore multiplies employment by about **0.6–0.8, not by 0.16**. Applied to the two curve calibrations at 15,000 sq ft (0.8–2.9 FTE at average second-home usage), the month-a-year

estate lands at **0.5–2.0 FTE, centrally about one full-time worker** — a person employed every year by a house that is empty 92% of the time.³

6. The labor balance: the statistic that matters

Housing policy needs a single honest number per unit: **workers supplied minus workers consumed**. A unit that houses a local worker adds labor to the economy; every unit also subtracts labor through the service employment it requires. The balance:

Unit	Workers housed	Workers consumed	Net labor balance
Studio, local worker	1.0 (fills ~1.2 jobs)	~0.01	≈ +1.2 jobs filled
Local family home (2 workers)	~1.7	~0.05	≈ +2.0 jobs filled
6,000 sq ft second home	0	0.2–0.7	≈ –0.2 to –0.7
15,000 sq ft, 1 month/yr	0	0.5–2.0	≈ –1 worker

Read down the last column and the Region’s arithmetic becomes visible: **each large, lightly-used home erases the workforce contribution of approximately one occupied studio** — and the worker it consumes must themselves be housed, fed, and schooled, which is the feedback loop of Section 8. This is also the honest foundation for mitigation policy: the products at the top of the curve generate five to a hundred times the employment of the products at the bottom, while today’s fee schedules — calibrated in 2005 — charge them a few thousand dollars against a liability that now prices in the hundreds of thousands.¹³

7. Construction: a Montrose flow, not a Telluride stock

Building a 15,000-square-foot custom home takes roughly **18–66 FTE-years of construction labor** over about two years — a crew averaging sixteen workers on site, by the 2000 survey’s San Miguel measurements.⁸ The old fee studies converted that one-time effort into a permanent local housing need by amortizing it over a 30–40-year “career” of a locally resident worker. The residency assumption was false when it was made: the 2000 survey itself found **51% of general contractors’ employees and 80% of subcontractors’ employees living outside San Miguel County**, with builders running van service for commuting crews.⁴ It remains false today: the construction workforce lives largely in Montrose and the West End, van-pools to job sites, and shows no intention of relocating to a county where the median home costs twenty-plus years of construction wages.

The honest accounting is this: construction employment is a **flow that follows construction volume**, wherever the volume goes. Its housing demand lands mostly in Montrose (where the

workers already live); what lands here is **traffic** — the van pools and pickups crossing Society Turn in the morning peak — plus midday commerce and staging demand. Roughly a fifth to a third of construction labor is locally resident,⁴ and only that share, only while building continues, belongs in a local housing-need calculation. A house should be charged for the permanent employment it creates every year of its life, not for a temporary workforce that was never going to live here — and the region’s construction pipeline (the hotels, Society Turn, the new treatment plant) should be evaluated for its cumulative traffic load on exactly the same reasoning.

8. The feedback loop, closed

Service workers need homes; occupied homes generate more service work. The loop is real and it converges. Each housed worker household generates roughly 0.2–0.4 local jobs (its own modest upkeep plus its local spending); each local job converts back to about half a housed household (workers hold ~1.2 jobs; households average 1.5–1.8 workers). The feedback ratio is therefore about 0.10–0.20, and the total need created by housing one worker household is a geometric series summing to **1.10–1.35 households** — a 10–35% markup, not a spiral.⁹ This independently reproduces the service-population overhead in the main report by an entirely different method, which is the kind of agreement that makes a number safe to use in public.

9. What this means

- **Mitigation should follow the measured curve.** Near zero at the bottom of the ladder; several-fold today’s fees at the top; occupancy-adjusted, because a month-a-year estate still runs 60–80% of a half-year home’s employment. The current schedule’s top brackets rest on an extrapolation the data contradict.⁵
- **Price the liability at today’s costs.** One FTE consumed is ~0.5 worker households to house; at the Region’s demonstrated public delivery costs (\$145,000–\$700,000 per unit at the relevant income tiers), the 15,000 sq ft estate embeds roughly **\$40,000–\$700,000 of public housing capital, centrally \$150,000–\$350,000** — versus an impact fee designed in 2005 dollars.¹³
- **Credit the labor balance.** A deed-restricted unit occupied by a local worker is the only product on the ladder with a positive balance; policy should treat it as workforce infrastructure, not as a generator of further need — its self-induced demand is the smallest number in this report.
- **Measure locally, once, properly.** Five survey questions replicating the 2013 Teton instrument — who maintains your home, how many hours, present and absent — would give this Region its first locally measured generation rates since 2000. Livable Telluride’s survey can carry them.¹⁴

Endnotes

1. Residential Job Generation Study (The Housing Collaborative, LLC / RRC Associates, December 2000), Part II, Tables II-10, II-11, II-13: total combined ongoing employment 0.23 FTE/unit overall; 0.29 San Miguel County; by total size: <3,000 sq ft 0.13 (n=1,814); 3,001–4,000: 0.16 (385); 4,001–5,000: 0.29 (261); 5,001–6,000: 0.42 (120); 6,001–8,000: 0.90 (75); >8,000: 1.10 (44). Our refit: ordinary least squares on $\ln(\text{FTE})$ vs $\ln(\text{size})$ across the six bin midpoints yields elasticity 1.57 (residual SD 0.30 in logs); an n-weighted fit gives 1.13, pulled by the heavily-sampled small-home bin whose composition skews to second-home condominiums.
2. Employee Generation by Land Use Study (Clarion Associates, Dr. James C. Nicholas, RRC Associates, for Teton County/Jackson, August 2013 public review draft), Tables 12–15: 978 survey responses; 87 FTE reported across 648 answering households (0.134 FTE/unit average); regression $\text{Employment} = \exp(-14.17 - 0.65 \cdot \text{Local} - 1.32 \cdot \text{SingleFamily} + 1.59 \cdot \ln(\text{sq ft}))$, $R^2 = 0.32$, all t-ratios ≥ 4.8 ; the authors cut predictions off at 7,000 sq ft for insufficient data beyond.
3. [E] 15,000 sq ft, non-locally-owned house: Teton calibration $\exp(-15.49 + 1.59 \cdot \ln 15,000) = 0.82$ FTE at average second-home usage; 2000-survey calibration (power fit at San Miguel level) ≈ 2.9 FTE. Both are extrapolations $\sim 65\%$ beyond the largest measured homes. Applying the occupancy factor of note 11 ($\times 0.59\text{--}0.81$, central 0.64) gives 0.5–2.0 FTE; central estimate $\approx 1.0\text{--}1.2$.
4. 2000 study, “Where Construction Workers Live” (San Miguel County): general contractors’ employees 23% Telluride region, 26% down-valley/mesas, 51% out of county; subcontractors’ employees 80% out of county; 18% of directly-reported employees were temporary residents; builders reported providing van service for commuting crews. Current-day van pools from Montrose: Livable Telluride direct observation, 2026 [C]. Local-resident share of construction labor used here: $\sim 20\text{--}30\%$.
5. Employee Housing Impact Fee Support Study (RPI Consulting for San Miguel County, December 2005 update): ongoing-employment curve $Y = 0.070174 \cdot e^{(0.000322 \cdot \text{sqft})}$, fitted to the 2000 study’s six bins and extended to 14,000 sq ft, where it predicts 6.36 FTE — 5.8 $\times$ the largest average ever measured (1.10 FTE, n=44); at 15,000 sq ft it predicts 8.8 FTE. Construction converted at 4.4 FTE-yr/1,000 sq ft $\div$ 40-year career; mitigation rate 37% set to the existing level of service (1,790 deed-restricted-housed employees $\div$ 4,777 R-1 jobholders); subsidy \$42,200/employee from six 1997–2005 projects (\$67,500/unit $\div$ 1.6 employees/unit).
6. 2000 study sampling: all local year-round residences under 3,000 sq ft were excluded from the homeowner survey frame (“none of the jurisdictions ... expressed a desire to target small to mid-range homes occupied as primary residences”); the San Miguel construction sample comprised 12 custom homes averaging 7,784 sq ft (87.5% over \$300/sq ft) plus one condominium unit, and the study warns it “should not be used to describe the overall construction industry.”
7. [E] Level reconciliation: the ratio of the 2000 overall average to the 2013 average ($0.23/0.134 \approx 1.7$) reflects sample design (note 6), the shift of labor into property-management firms, and

regional differences. San Miguel's measured premium over the four-area average ($0.29/0.23 = 1.26$) is applied to the high calibration. The truth for today's San Miguel County most likely lies between the two curves; the band in Exhibit A is that statement drawn honestly.

8. Construction intensity: 2013 Teton study, ES-202 payroll employment in building construction $\div$ ten years of permitted floor area = 1.234 FTE-yr/1,000 sq ft (administrative data; excludes sole proprietors and 1099 labor). 2000 study: Method 1 (contractor headcount observation) 2.3; Method 2 (contract dollars $\div$ trade labor shares $\div$ wages) 4.4. A 15,000 sq ft home at 1.2–4.4 FTE-yr/1,000 sq ft = 18–66 FTE-years; the 2000 study measured an average crew of 16 on San Miguel custom homes and ~20-month build times.

9. [E] Loop arithmetic: jobs generated per housed worker household $g \approx 0.2\text{--}0.4/\text{yr}$ (own-unit upkeep 0.03–0.10 + local spending 0.15–0.30); jobs convert to housed households at $\div 1.2$ jobs/worker $\div 1.5\text{--}1.8$ workers/household $\approx \times 0.5$; feedback ratio $r = 0.10\text{--}0.20$; multiplier $1/(1-r) = 1.10\text{--}1.25$ (≈ 1.35 at the pessimistic edge). Matches the 15–35% service-population overhead derived from peer-community benchmarks in Housing in the Upper San Miguel, Section 7.

10. Demand-side channel: National Association of Home Builders estimates ~30 ongoing local jobs per 100 occupied affordable units ($\sim 0.30/\text{unit}$) from resident spending (as reported in The Role of Affordable Housing in Creating Jobs and Stimulating Local Economic Development, Center for Housing Policy, January 2011). Discounted here to 0.15–0.30 for down-valley and online retail leakage [E]. Second-home spending while present is largely captured in commercial (restaurant/retail) employment generation and is credited only a sliver here to avoid double-counting.

11. 2000 study, Table II-7 (employee hours when owner present vs absent): housekeepers 7.4 vs 2.5 hrs/wk (34%); gardeners/exterior 6.9 vs 2.4 (35%); caretakers 12.3 vs 7.6 (62%). Occupancy factor for a 4-week home relative to the 25.5-week average second home: $(4 + 48 \cdot a)/(25.5 + 26.5 \cdot a)$ where $a =$ absent-week intensity 0.34–0.62 $\rightarrow$ factor 0.59–0.81, central 0.64 at $a = 0.4$. [E]

12. 2000 study, Table II-4: second homes used 25.5 weeks/year on average across all purposes (9.2 owner weeks, 7.7 vacation-rental, 5.4 long-term rental, 3.0 primary), vacant 26.5 weeks (51%).

13. [E] Liability pricing: 0.5–2.0 FTE consumed $\times \sim 0.5$ housed households per job $\times$ loop multiplier 1.1–1.35 $\approx 0.3\text{--}1.3$ worker households; at the Region's demonstrated public delivery gap of \$145,000–\$700,000 per unit at 60–100% AMI (Housing in the Upper San Miguel, Sections 3 and 5), $\approx$ \$40,000–\$700,000+; central: 0.5 households $\times$ \$300,000–\$700,000 $\approx$ \$150,000–\$350,000. Compare the 2005-calibrated fee schedule: $\sim$ \$7,200–\$9,600 at 5,000–6,000 sq ft (2011 column).

14. The 2013 Teton instrument's employment battery (direct hires, property-management, HOA, caretaker, other services; hours present/absent) is the survey-ready template. A San Miguel replication would also resolve the level question of note 7 with current local data.

— *Draft for discussion. Livable Telluride welcomes corrections; the analysis improves when readers find its errors.*